

Jay Niketan Pathare

jayadmit456@gmail.com | +1(716)303-9924 | NY, US | Portfolio | LinkedIn | GitHub

EDUCATION

Master's in Computer Science

University at Buffalo, The State University of New York, Buffalo, NY

August 2025 – December 2026

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society's Institute of Technology, Mumbai University, India

August 2019 – May 2023

EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 5 | HeinOnline, Buffalo

January 2026 – April 2026

- Engineered a fault-tolerant OCR-LLM pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC metadata from 47 law journals (28,167+ pages).
- Built a fuzzy-matching combinator engine to merge asynchronous JSON outputs into a validated YAML structure, cutting manual data-entry workload by ~90% and raising extraction accuracy from 57.03% to 90.81% over five iterations (43 of 47 journals $\geq 80\%$, 8 at 100%; validated against ground truth).
- Parallelized the pipeline via a worker pool (cores – 1) that pulls the next journal from a shared queue on completion, and added an "aiscore" system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Led delivery of platform modules end to end: gathered requirements in client meetings, built prototypes to validate scope, decomposed the finalized work across a small team, and reviewed and merged their contributions into production.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
- Delivered a telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation.
- Integrated a color-blindness diagnostic module (Ishihara plate testing, ~97–99% clinical sensitivity) with automated reporting and real-time result visualization for practitioners.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups, ensuring secure, low-latency communication for backend services.
- Automated cloud resource provisioning, reducing setup time by ~40% and minimizing deployment-related issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, generating reports highlighting key operational trends.

TECHNICAL SKILLS

- Languages & Databases:** Python, C, SQL, MySQL, PostgreSQL, HTML/CSS
- Frameworks:** Django, Django REST Framework, Django Channels, ReactJS, Pandas, NumPy, Matplotlib
- AI/ML:** PyTorch, PyTorch Geometric, scikit-learn, TensorFlow, GraphSAGE, GAT, CNNs, RNNs, GNNs, LLM Integration, Prompt Engineering
- Software Engineering:** OOP, REST APIs, Full-Stack Development, Backend Development, System Design, ETL, Unit Testing, Load Testing, Backtracking Algorithms, Agile, Real-Time Monitoring
- Tools:** Linux, Docker, Docker Compose, Redis, AWS (EC2/RDS), Git, GitHub, GitHub Copilot, OAuth/Auth0, WebRTC, WebSockets, Locust, Tableau

ACADEMIC PROJECTS

Decoupled Learning Management System (LMS) | Individual Project

July 2026

- Architected a full-stack LMS (Django REST Framework, React) with Redis-backed real-time messaging via Django Channels for live quizzes, leaderboards, and rate-limited chat; served 45+ concurrent users with sub-500ms latency.
- Implemented idempotent server-side quiz grading, role-based access control for instructors/students, and a backtracking algorithm to generate conflict-free course schedules from section and term constraints.
- Secured the LMS with stateless JWT and custom WebSocket authentication; load-tested with Locust and asyncio, cutting p95 REST latency 26% and quiz-submission latency from 2000ms to 1500ms.

Fraud Detection using Graph Neural Networks | Individual Project | University at Buffalo, NY

March 2026

- Developed a GraphSAGE-based fraud detection model on the IEEE-CIS dataset (590k+ records), using Focal Loss to handle a severe 27:1 class imbalance and achieving a 0.9259 AUC-ROC (vs. 0.8584 MLP baseline, 0.7928 GAT).
- Converted tabular transaction data into a graph network using shared card, device, and email identifiers, capping node groups to prevent memory explosion while capturing relational fraud rings.

Pintos Operating System – User Programs Layer | Individual Project, University at Buffalo

May 2026

- Engineered the user programs layer for Pintos, a teaching x86 OS kernel in C, designing the process lifecycle: stack setup, execution, parent-child synchronization, and clean termination.
- Developed a system call handler supporting 13 operations for process control and file I/O, with thread-safe file descriptor management for 128 open files per process and executable write-denial protection.
- Solved concurrency via heap-allocated synchronization primitives, passing 100% of 80 test cases.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Cloud Practitioner** July 2024
- Certificate of Appreciation:** CSE Workshop "GitHub: Hands-On from Basics to Advanced," UB February 2026
- Research Publication:** J. Pathare et al., "Music Genre Classification," IJRAR, Apr. 2023 (97.68% accuracy). [Link](#).